



FRONTEND DEVELOPMENT - I

CSC 1051: FULL-STACK FUNDAMENTALS LABORATORY

N.H.Wanigasingha (Mphil (Reading), BSc (Hons))

Department of Computer Science

Faculty of Applied Sciences

University of Sri Jayewardenepura

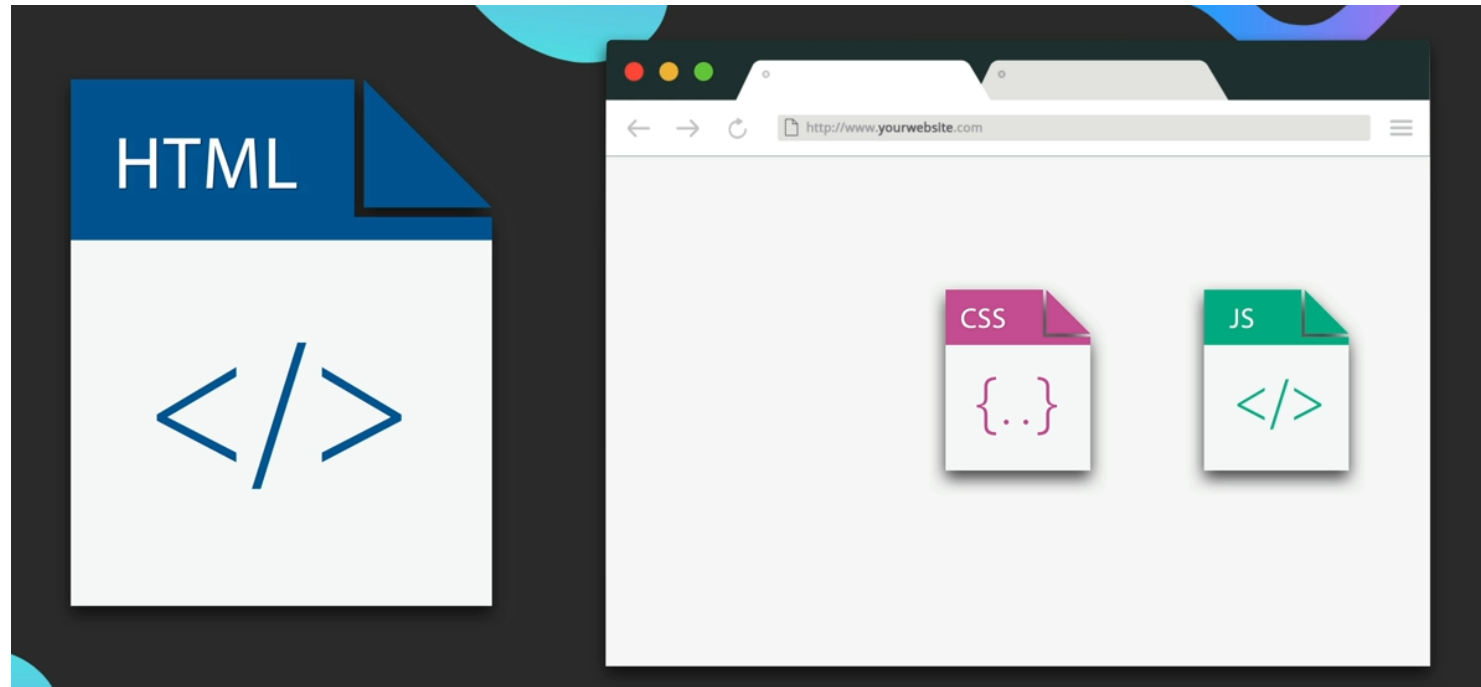
FRONTEND DEVELOPMENT

- Front-end Development is the development or creation of a user interface using some markup languages and other tools.
- It is the development of the user side where only user interaction will be counted.
- It consists of the interface where buttons, texts, alignments, etc are involved and used by the user.
- **Languages for Frontend Development:**
 - **HTML** : supports forms, tables, and other elements needed to organize information, and continues to evolve to help developers build interactive, accessible interfaces across platforms.
 - **CSS**: complements HTML by adding styling, managing colors, and enabling responsive designs. CSS helps developers create visually appealing, consistent layouts across devices and screen sizes.
 - **JavaScript**: enables dynamic updates and event handling in web development, making it key for building interactive and responsive user interfaces.

INTRODUCTION TO HTML

- HTML is a language which is used to create document on the World Wide Web.
- HTML stands for **H**yper **T**ext **M**arkup **L**anguage
- HTML is not a programming language, it is **a markup language** which read and interpreted by your browser
- A markup language is a set of **markup tags**
- HTML uses **markup tags** (HTML Tags) to describe web pages that run on browser

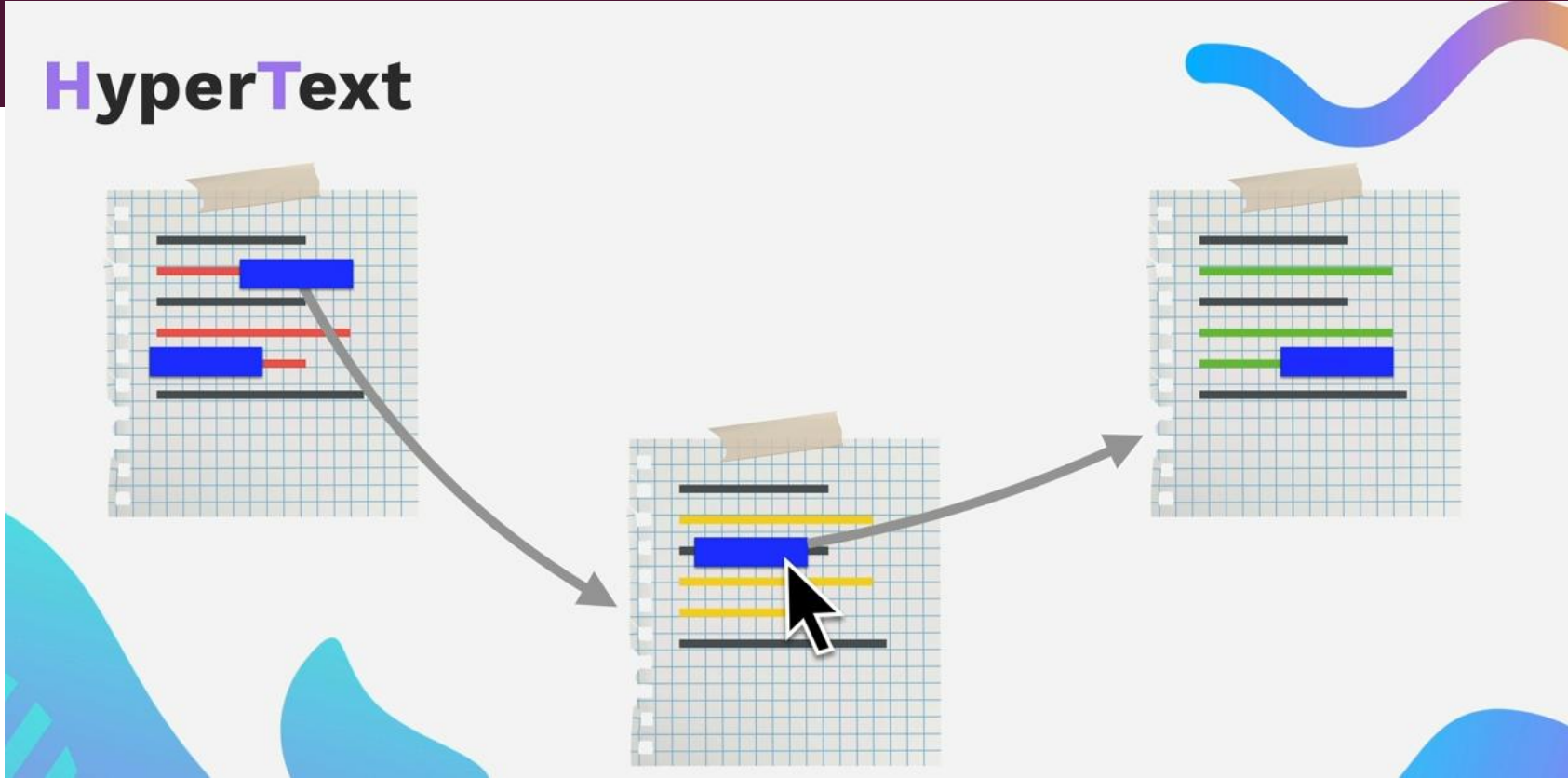
HTML, CSS, JS



HTML Defines the content and structure of the website

HTML

HyperText



Refers to the pieces of text which can link to other documents in the website

SAMPLE HTML WEBSITE

<https://www.w3.org/History.html>

HTML

Markup Language

“This is a quote”



What is a markup language?

The presence of these quotation marks is what tells a reader that this part is a quotation.

ALL HTML5 TAGS

<a>	<canvas>	<dt>	<iframe>	<meta>	<rp>	<sup>
<abbr>	<caption>			<meter>	<rt>	<svg>
<address>	<cite>	<embed>	<input>	<nav>	<ruby>	<table>
<area>	<code>	<fieldset>	<ins>	<noscript>	<s>	<tbody>
<article>	<col>	<figcaption>	<kbd>	<object>	<samp>	<td>
<aside>	<colgroup>	<figure>	<keygen>		<script>	<template>
<audio>	<data>	<footer>	<label>	<optgroup>	<section>	<textarea>
	<datalist>	<form>	<legend>	<option>	<select>	<tfoot>
<base>	<dd>	<head>		<output>	<small>	<th>
<bdi>		<header>	<link>	<p>	<source>	<thead>
<bdo>	<details>	<hgroup>	<main>	<param>		<title>
<blockquote>	<dfn>	<h1> to <h6>	<map>	<picture>		<tr>
<body>	<dialog>	<hr>	<mark>	<pre>	<style>	
 	<div>	<html>	<menu>	<progress>	<sub>	<var>
<button>	<dl>	<i>	<menuitem>	<q>	<summary>	<video>
						<wbr>

IMPORTANT ONES

<a>	<canvas>	<dt>	<iframe>	<meta>	<rp>	<sup>
<abbr>	<caption>			<meter>	<rt>	<svg>
<address>	<cite>	<embed>	<input>	<nav>	<ruby>	<table>
<area>	<code>	<fieldset>	<ins>	<noscript>	<s>	<tbody>
<article>	<col>	<figcaption>	<kbd>	<object>	<samp>	<td>
<aside>	<colgroup>	<figure>	<keygen>		<script>	<template>
<audio>	<data>	<footer>	<label>	<optgroup>	<section>	<textarea>
	<datalist>	<form>	<legend>	<option>	<select>	<tfoot>
<base>	<dd>	<head>		<output>	<small>	<th>
<bdi>		<header>	<link>	<p>	<source>	<thead>
<bdo>	<details>	<hgroup>	<main>	<param>		<title>
<blockquote>	<dfn>	<h1> to <h6>	<map>	<picture>		<tr>
<body>	<dialog>	<hr>	<mark>	<pre>	<style>	
 	<div>	<html>	<menu>	<progress>	<sub>	<var>
<button>	<dl>	<i>	<menuitem>	<q>	<summary>	<video>
						<wbr>

TAGS VS ELEMENTS

<Tag> vs. Element



THE HEADING TAG

```
<h1>Heading 1</h1>
```

```
<h2>Heading 2</h2>
```

```
<h3>Heading 3</h3>
```

```
<h4>Heading 4</h4>
```

```
<h5>Heading 5</h5>
```

```
<h6>Heading 6</h6>
```

Heading 1

Heading 2

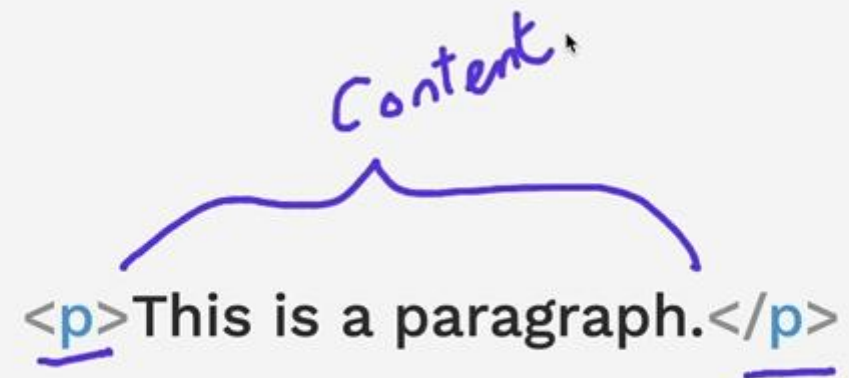
Heading 3

Heading 4

Heading 5

Heading 6

THE PARAGRAPH TAG



<p>This is a paragraph.</p>

The diagram illustrates the structure of a paragraph tag in HTML. It shows the opening tag <p> and the closing tag </p>, both underlined. A blue bracket spans the text "This is a paragraph." between the two tags. Above the bracket, the word "Content" is written in a handwritten style, indicating that the text between the tags is the content of the paragraph.

THE PARAGRAPH TAG

```
<p>This is a paragraph</p>  
<p>This is a paragraph</p>
```

This is a paragraph

This is a paragraph

Task: Break a sample text into paragraphs

VOID ELEMENTS / SELF CLOSING TAGS

No content in the Tag!!



```
<p>This is a paragraph.</p>  
<hr />
```

The diagram shows two lines of HTML code. The first line is `<p>This is a paragraph.</p>` and the second line is `<hr />`. In both lines, a blue arrow points to the closing slash (`/`) of the tag, indicating that these are self-closing tags and do not require a separate closing tag.

`<hr />` - Horizontal Rule element

`<br />` - break element

HORIZONTAL RULE AND BREAK ELEMENT

```
<p>This is a paragraph</p>  
<hr />  
<p>This is a paragraph</p>
```

This is a paragraph

This is a paragraph

```
<p>  
To see a World in a Grain of Sand<br />  
And a Heaven in a Wild Flower,<br />  
Hold Infinity in the palm of your hand<br />  
And Eternity in an hour.<br />  
</p>
```

To see a World in a Grain of Sand
And a Heaven in a Wild Flower,
Hold Infinity in the palm of your hand
And Eternity in an hour.

LIST ELEMENTS

Unordered List + List Items

```
<ul>  
  <li>Milk</li>  
  <li>Eggs</li>  
  <li>Flour</li>  
</ul>
```

Ordered List + List Items

```
<ol>  
  <li>Milk</li>  
  <li>Eggs</li>  
  <li>Flour</li>  
</ol>
```

NESTING AND INDENTATION

```
<ul>
  <li>Wake up and brush teeth</li>
  <li>Drink 500ml warm water</li>
  <li>Do yoga for an hour</li>
  <li>Make omelette
    <ul>
      <li>Whisk eggs with milk</li>
      <li>Add butter to pan</li>
      <li>Add in eggs and stir</li>
      <li>When solid add salt</li>
    </ul>
  </li>
  <li>Start work</li>
</ul>
```

- Wake up and brush teeth
- Drink 500ml warm water
- Do yoga for an hour
- Make omelette
 - Whisk eggs with milk
 - Add butter to pan
 - Add in eggs and stir
 - When solid add salt
- Start work

NESTING AND INDENTATION

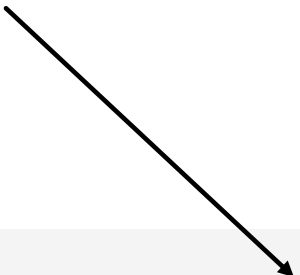
```
<ul>
<li>Wake up and brush teeth</li>
<li>Drink 500ml warm water</li>
<li>Do yoga for an hour</li>
<li>Make omelette
  <ul>
    <li>Whisk eggs with milk</li>
    <li>Add butter to pan</li>
    <li>Add in eggs and stir</li>
    <li>When solid add salt</li>
  </ul>
</li>
<li>Start work</li>
</ul>
```

Indentation is important!!!

ANCHOR ELEMENTS

HTML Attribute

```
<a>This is a link</a>
```



```
<a href="http://www.google.com">This is a link</a>
```

It enables us to create hyperlinks

HTML ATTRIBUTES

Global attribute → can be used with many/all HTML elements

Specific attribute → designed for a particular HTML element

Global attributes are attributes that can be set on any HTML element, including elements in the `<head>`. There are more than 30 global attributes.

```
<a draggable=true  
This is a link to Google  
</a>
```

HTML DOCUMENTATION

<https://developer.mozilla.org/en-US/docs/Web/HTML>

IMAGE ELEMENT

```

```

```

```

alt attribute stands for Alternative Text Description

A screen reader will read out the alt text when it hits an image

FILE PATHS

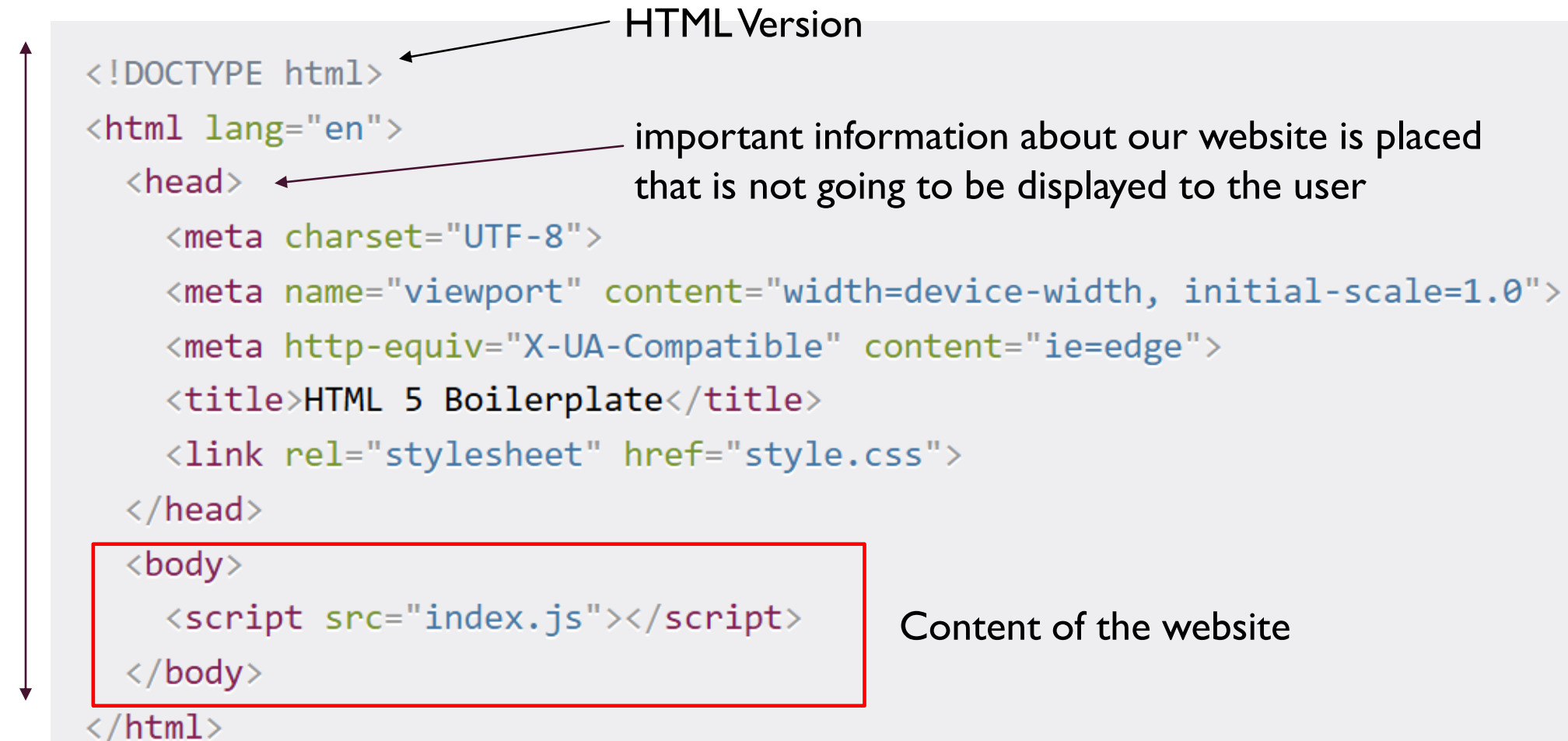
```
C:/Project/Images/cat.png
```

Absolute file paths

```
/Project/Images/cat.png
```

Relative file paths

PROPER HTML STRUCTURE



The diagram illustrates the proper HTML structure with the following code and annotations:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta http-equiv="X-UA-Compatible" content="ie=edge">
    <title>HTML 5 Boilerplate</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <script src="index.js"></script>
  </body>
</html>
```

Annotations:

- HTML Version**: Points to the `<!DOCTYPE html>` declaration.
- important information about our website is placed that is not going to be displayed to the user**: Points to the `<head>` section.
- Content of the website**: Points to the `<body>` section, which is highlighted with a red box.

A vertical double-headed arrow on the left side of the code block indicates the overall structure of the document.

HTML LAB EXERCISE: CREATE A PERSONAL PORTFOLIO WEBPAGE

Instructions:

1. Webpage Title: Set the title of the webpage as "My Portfolio."

2. Header Section:

Add a heading (`<h1>`) with your name. Include a short introduction or tagline below the heading using a `<p>` tag.

3. About Me Section:

Create a section with a heading (`<h2>`) titled "About Me." Write a short paragraph about yourself.

Skills Section:

Use an unordered list (`<ul>`) to list at least 5 of your skills.

5. Projects Section:

Add a heading (`<h2>`) titled "Projects."

Use a table (`<table>`) to display at least 3 projects. Include columns for:

Project Name

Description

Technology Used

HTML LAB EXERCISE: CREATE A PERSONAL PORTFOLIO WEBPAGE

6. Contact Section:

Add a heading (<h2>) titled "Contact Me."

Create a contact form with:

- Text field for "Name"

- Email field

- Message text area

- Submit button

7. Footer:

Add a footer with your social media links (e.g., GitHub, LinkedIn) as clickable hyperlinks.